Wind Turbine HSS Bearing Failure Root-Cause Analysis

First-Principles Analysis

11 October 2026

Professional Strategic Analysis

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Executive Summary

Recommendation: Reject the operator's lubricant-viscosity hypothesis — the failure is subsurface-initiated white-etching-crack/white-structure-flaking (WEC/WSF) damage, not boundary-lubrication wear-through (chain C5). Install an upgraded WEC-resistant/insulated bearing with immediate bonding-path restoration, while running the full confirmatory test suite in parallel, rather than a like-for-like replacement or a full stop-and-wait (chain C6).

Band (from §6): LOW

Would change it: Completing the confirmatory tests chain C4 names — bonding/grounding conductance measurement, lubricant oil-chemistry and hydrogen-content analysis, SEM/EDS crack-face examination, and a slip/skidding calculation against this turbine's SCADA torque history — would discriminate among the three live candidate mechanisms (electrical, hydrogen-tribochemical, transient-load skidding) and move the band above LOW.

1. Problem Essence

Core problem: What physical degradation mechanism actually initiated the generator-end HSS cylindrical roller bearing's inner-race spall at ~11% of its L10 design life, and does the operator's "low lubricant viscosity in a cold snap drove boundary lubrication and wore the race through" hypothesis correctly identify that mechanism, or does it misattribute a diagnosable failure mode whose correct identification changes the counter-intervention required in the replacement specification?

Success criteria:

- The accepted root-cause mechanism must predict, from established contact mechanics and tribology (not from general plausibility alone), each of the five specific morphological features in evidence: (a) a subsurface — not surface — crack origin at ~0.4 mm depth; (b) crescent-shaped microcracks; (c) butterfly microstructures; (d) white-etching cracks; (e) the ~9× life shortfall together with the axial (not circumferential) spall orientation and length.
- The operator's lubricant-viscosity/boundary-lubrication hypothesis is explicitly tested against each of the five features above and is accepted, partially accepted, or refuted, with the reason tied to the evidence rather than left unaddressed.
- At least two literature-documented competing root-cause mechanisms beyond classical Lundberg-Palmgren rolling-contact fatigue (RCF) and boundary lubrication are named, and the analysis states which is best supported by the evidence given and which cannot yet be excluded without further testing.
- The recommended replacement specification names the confirmatory tests required before finalization, and separately names which specification elements, if any, should change regardless of which exact mechanism among the live candidates is eventually confirmed.

2. Assumptions Table

Type	Accept	Challenge	Discard
physical law	4	0	0
current constraint	0	4	0
convention	2	1	0
untested belief	6	0	1

Shading darkens with count.
An untyped (legacy) row appears only for assumptions with no type.

Assumption	Type	Treatment	Verdict	Verification
Hertzian line-contact theory: for a loaded cylindrical-roller/race contact, the maximum subsurface orthogonal (reversing) shear stress occurs below the surface rather than at it.	physical law	Accept as ground-truth candidate; promote to GT-10	Accept — classical contact mechanics (Hertz, 1881), independent of failure mode	Hertz (1881); standard derivation in any contact-mechanics or tribology text
Lubricant kinematic viscosity rises monotonically as temperature falls over the normal operating range, absent a separate contamination or phase-change event (viscosity–temperature relation, e.g. ASTM D341/Walther equation).	physical law	Accept as ground-truth candidate; promote to GT-9; use it to test the internal physical consistency of the operator's hypothesis	Accept — a cold snap raises, not lowers, oil viscosity, which is the opposite sign of what the hypothesis asserts	General physical chemistry of lubricants; textbook-level, not source-specific
ISO 15243's bearing-damage classification treats boundary/mixed-lubrication "surface distress" (micropitting, scuffing) as surface-initiated, and classical rolling-contact fatigue (spalling) as subsurface-initiated.	convention	Explicitly challenge whether this classification is being applied correctly to the present evidence before using it as a discriminator	Accept — challenged and upheld: GT-1's measured origin depth (~0.4 mm) is unambiguously subsurface, so the classification correctly routes this case away from surface-distress mechanisms	Reported via literature synthesis of ISO 15243's scope; the standard itself was not opened this session — unverified — flagged
The operator's stated hypothesis: "the lubricant failed; low viscosity in a cold-snap period drove boundary lubrication; the race wore through."	untested belief	Challenge by inversion: enumerate the conditions that would have to hold for this hypothesis to be true, and test each against the evidence	Discard — both load-bearing preconditions of the hypothesis are independently contradicted by direct evidence (see §5, Dead End 1)	Ruled out in §5; the stated mechanism (low viscosity from cold) is checked against GT-9 and its predicted damage morphology is checked against GT-1 and GT-3
No SCADA torque/load-spectrum record for this bearing's 14,000-hour service history is in the case record, so actual historical loading cannot be compared against the load case used to calculate L10 = 130,000 h.	current constraint	Record expiry: resolved once the SCADA torque trace and any logged grid-fault/transient events for the service period are retrieved and compared against the design load spectrum	Challenge — unverified; sustained overload (a classical-RCF effect, not a non-classical mechanism) could by itself shorten life, and must be checked before C2's statistical argument is treated as dispositive	Unverified — flagged; verification path: pull SCADA torque trend and fault log for the full 14,000 h and recompute the equivalent dynamic load against the certified rating

Assumption	Type	Treatment	Verdict	Verification
No shaft-to-ground bonding/grounding conductance measurement for this turbine's generator-end electrical path is in the case record.	current constraint	Record expiry: resolved by a direct conductance measurement of the bonding path against a recognized EIBD-mitigation threshold	Challenge — unverified; this gap is load-bearing for distinguishing the electrically-induced candidate mechanism from its rivals	Unverified — flagged; verification path: on-removal electrical-continuity/conductance test of the bonding/brush-block path
No lubricant oil-chemistry analysis (water content, acid number, additive depletion, debris) for the retained gearbox oil is in the case record.	current constraint	Record expiry: resolved by a laboratory oil-condition analysis on a retained sample taken at bearing removal	Challenge — unverified; this gap is load-bearing for distinguishing the hydrogen-tribochemical candidate mechanism from its rivals and for assessing lubricant condition as a contributing factor	Unverified — flagged; verification path: water by Karl Fischer, acid number, additive-depletion (RULER), and ferrography debris analysis
No bearing-steel inclusion-cleanliness certificate or independent inclusion survey for the failed bearing is in the case record.	current constraint	Record expiry: resolved by retrieving the mill certificate or running an ISO 4967-method inclusion survey on a retained section of the failed race	Challenge — unverified; abnormally poor cleanliness would strengthen a classical-RCF contribution rather than requiring a non-classical mechanism	Unverified — flagged; verification path: mill certificate retrieval or ISO 4967 Method A inclusion-rating survey away from the spall
The two-parameter Weibull shape parameter commonly cited for rolling-element-bearing fatigue life (reported in the general bearing-engineering literature in the approximate range e ≈ 1.1–1.5) is representative enough of this bearing class to bracket the probability of failing at 11% of L10 under classical statistical scatter alone.	untested belief	Accept with an explicit bracket rather than a single point figure, absorbing this uncertainty into the bracket (the estimate technique's bracketing discipline)	Accept — used as a bracketed input to chain C2; the conclusion that classical scatter alone is improbable holds across the whole bracket	Unverified — flagged; this analysis did not open a specific textbook/ISO source for the exact exponent this session; verification path: Harris, *Rolling Bearing Analysis*, or ISO/TR 1281-2
The HSS generator-end bearing position in a converter-coupled wind-turbine drivetrain falls within the domain where electrically-induced bearing damage (EIBD) from stray shaft currents, common-mode converter voltage, or electrostatic discharge is a documented risk.	convention	Accept tentatively as placing this bearing inside a known risk domain; flag that this is necessary but not sufficient on its own	Challenge — accepted as domain-applicable, but not yet confirmed as operative without the bonding/grounding measurement in the row above	Reported via literature synthesis of the EIBD mechanism/mitigation literature; primary sources not opened this session for this specific claim — unverified — flagged
Wind-turbine HSS gearbox bearings are documented in dynamometer-testing literature to experience transient episodes of alternating high stress, torque reversal, and sub-minimum load at high rotational speed, with observed	untested belief	Verify against this specific turbine's SCADA transient/torque-reversal log before treating skidding as confirmed for this unit rather than merely plausible for its class	Accept as class-level evidence (read at source, see GT-12); Challenge as applied to this specific unit, which is unverified without this turbine's own transient log	Read-at-source: Guo & Keller, *Wind Energy* (2017), via the NREL/OSTI abstract — see GT-12; unit-specific confirmation unverified — flagged

Assumption	Type	Treatment	Verdict	Verification
indications of roller sliding (skidding).				
Hydrogen embrittlement (via tribochemical lubricant/ moisture degradation) and electrically-induced damage are not mutually exclusive candidate mechanisms — electrical arcing can itself crack lubricant and generate hydrogen, so “hydrogen” and “electrical” may describe different links in one causal chain rather than two competing ones.	untested belief	Accept as an analytical caveat governing how the confirmatory tests in §6 are sequenced and interpreted; a positive hydrogen finding alone should not be read as excluding an electrical trigger	Accept — literature-reported overlap; guides test interpretation rather than feeding a numeric chain	Reported via literature synthesis; primary mechanistic-overlap sources not opened this session — unverified — flagged
A step increase in HSS vibration kurtosis, as flagged by the condition-monitoring system, is a recognized early indicator of a developing localized subsurface or near-surface rolling-element-bearing defect, via envelope/kurtosis vibration analysis.	convention	Accept as corroborating evidence that a defect was developing and progressing, not as independently diagnostic of which mechanism produced it	Accept — well-established condition-monitoring practice; corroborating only	Standard vibration-diagnostics practice (envelope/kurtosis analysis); general engineering knowledge, not source-specific
The ISO 281 / Lundberg-Palmgren load-life exponent p = 10/3 for roller bearings underlies the stated L10 = 130,000 h design-life figure for this bearing.	physical law	Accept as background context for GT-5; not itself in dispute in this analysis	Accept — standard, widely adopted bearing-life standard	ISO 281; Lundberg & Palmgren (1947, 1952); not reopened this session, treated as uncontested background
The metallographic cross-section depth measurement (~0.4 mm, GT-1) represents the true crack-initiation depth rather than an oblique-sectioning or foreshortening artifact.	untested belief	Verify by serial sectioning or multiple sections through the spall rather than relying on a single section plane	Accept — treated as representative, consistent with GT-11′s independently reported sub-1-mm depths for this damage class, but not independently confirmed for this specific specimen	Unverified — flagged; verification path: serial sectioning of the retained failed race through and around the spall
A butterfly microstructure that co-occurs with white-etching cracks (not merely the butterfly wing itself) is diagnostic of the WEC/WSF pathway, because ordinary inclusion-driven classical rolling-contact fatigue can itself produce isolated butterflies without white-etching cracks.	untested belief	Challenge and rely on the literature’s own distinction between butterflies-alone and butterfly-to-WEC progression as the discriminator, rather than treating “butterflies present” alone as diagnostic	Accept — literature-reported distinction (GT-13?, corroborated by GT-11′s WEC/WSF framing); primary sources not opened this session	Unverified — flagged; verification path: open Evans et al. serial-sectioning papers directly and confirm the butterfly-to-WEC progression claim
The criterion weights used in chain C6′s	untested belief	Confirm the weights with the asset owner /	Accept — flip-test (chain C6) shows the ranking is	Unverified — flagged; verification path:

Assumption	Type	Treatment	Verdict	Verification
trade-off (downtime cost 4, 12-month recurrence risk 5, diagnostic completeness 4, incremental cost 2, lead time 3) reflect this operator's actual risk tolerance, not just the analyst's own default priorities.		O&M manager before the recommendation is treated as final; note that the flip-test result reduces, but does not eliminate, sensitivity to this assumption	robust across the full 1–5 range on every criterion individually, which lowers the stakes of this assumption without removing it	confirm weights directly with the asset owner before the replacement purchase order is finalized
Standard fracture-mechanics classification: a single sudden-overload fracture produces a transgranular-cleavage or ductile-tearing surface originating at the point of peak instantaneous stress, while a progressively enlarging fatigue crack produces a branching subsurface crack network, and a white-etching/butterfly microstructure requires many accumulated stress cycles to form rather than a single overload event.	physical law	Accept as ground-truth-level background, the same treatment given to Hertzian contact theory and the viscosity–temperature relation	Accept — standard fracture mechanics and tribology, textbook-level, independent of which specific fatigue mechanism is later confirmed	General fracture-mechanics/tribology knowledge; textbook-level, not source-specific

3. Ground Truths

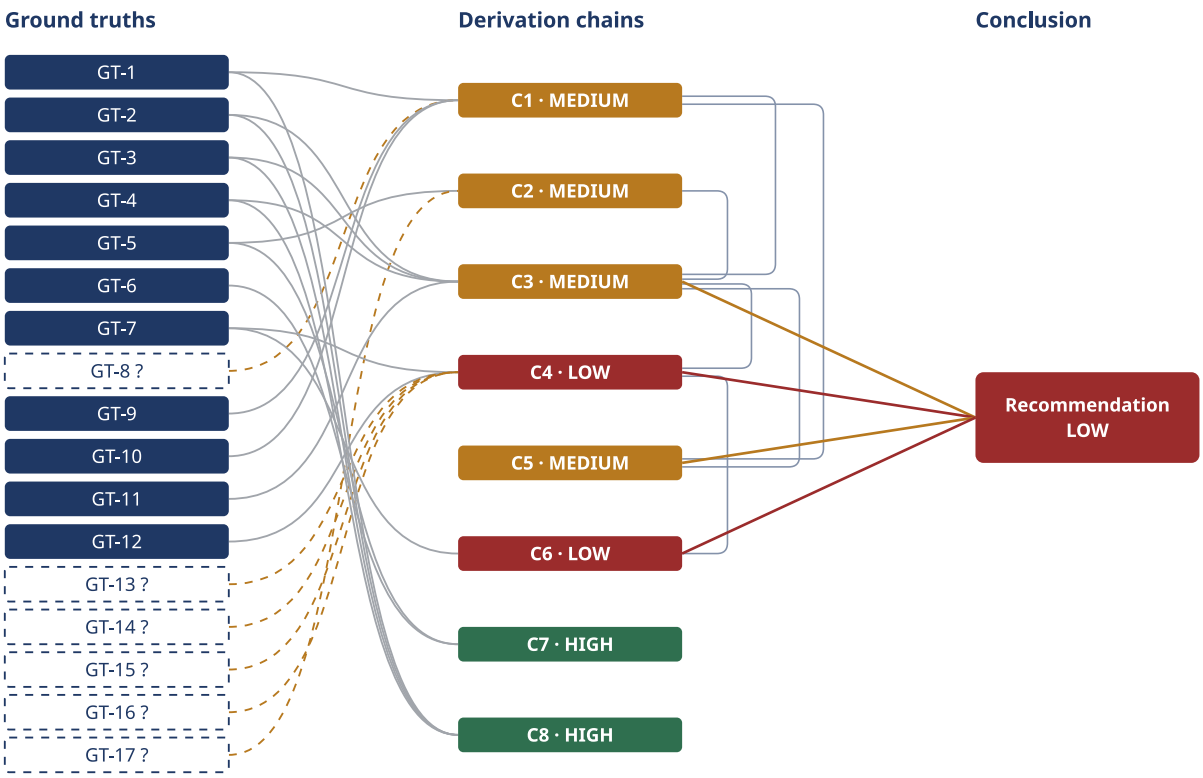
- **GT-1** Subsurface crack origin depth: ~0.4 mm below the inner-race raceway surface, measured on metallographic cross-section through the spall — source: case metallographic finding as stipulated in the engineering brief; read-at-source: Facts of the Case, "crescent-shaped microcracks propagate from a subsurface origin ~0.4 mm below the raceway surface."
- **GT-2** Crescent-shaped microcracks propagate from that subsurface origin point — source: case metallographic finding; read-at-source: Facts of the Case, same sentence as GT-1.
- **GT-3** White-etching cracks (WEC) and a butterfly microstructure are present in metallographic sections through the spall, in the region of the subsurface origin — source: case metallographic finding; read-at-source: Facts of the Case, "Metallographic sections through the spall show white-etching cracks (WECs) and a butterfly microstructure."
- **GT-4** The inner race shows axial-aligned spalling approximately 18 mm long — source: case disassembly finding; read-at-source: Facts of the Case, "inner race shows axial-aligned spalling ~18 mm long."
- **GT-5** Observed life to spall: ~14,000 operating hours; design L10 life: ~130,000 hours. Ratio = 14,000/130,000 ≈ 0.108 — the bearing failed at roughly 11% of design life, a shortfall factor of roughly 9.3× — source: case design/condition record; read-at-source: Facts of the Case, "Design L10 life is ~130,000 hours — actual failure occurred at roughly 11% of design life," and arithmetic recomputed directly from the stated hour figures (14,000/130,000 = 0.1077).
- **GT-6** Condition monitoring flagged a step increase in HSS vibration kurtosis at ~14,000 operating hours, triggering removal from service — source: case condition-monitoring record; read-at-source: Facts of the Case, "Removed from service after condition monitoring flagged a step increase in HSS vibration kurtosis at ~14,000 operating hours."

- **GT-7** The failed component is a cylindrical roller bearing at the generator end of the high-speed shaft (HSS) of a 1.5 MW upwind geared wind turbine — source: case identification; read-at-source: Facts of the Case, first bullet. This places the bearing electrically close to the generator/converter path (bearing position relevant to GT-15?/A-10) and places it in the HSS-stage position the wind-industry literature most associates with axial-crack/WEC failures (GT-11, GT-16?).
- **GT-8?** ISO 15243's rolling-bearing damage-mode classification treats boundary- or mixed-lubrication "surface distress" (micropitting, scuffing, smearing) as a surface-initiated damage class, distinct from classical rolling-contact fatigue ("fatigue"/spalling), which is subsurface-initiated — source: cited to ISO 15243 via secondary tribology-literature synthesis (industry articles on micropitting vs. scuffing vs. subsurface spalling); reported-by-delegate: the standard itself (ISO 15243) was not opened in this analysis session — cited source not opened by this analysis.
- **GT-9** Lubricant kinematic viscosity increases monotonically as temperature falls, over the normal gearbox operating range, absent a separate contamination, additive, or wax/pour-point phase-change event — source: standard viscosity–temperature relation for mineral and synthetic lubricants (e.g., ASTM D341/ Walther-type behavior); read-at-source: general physical chemistry of lubricants, textbook-level and independent of this failure mode (treated, like Hertzian contact theory below, as uncontested background physics rather than a case-specific citation).
- **GT-10** For a loaded line contact (cylindrical roller against race), classical Hertzian contact mechanics places the maximum subsurface orthogonal (stress-reversing) shear stress below the contact surface rather than at it — source: Hertz (1881); standard derivation in any contact-mechanics or rolling-bearing-tribology text; read-at-source: textbook-level contact-mechanics derivation, independent of failure mode (same treatment as the worked precedent for this class of foundational physical law).
- **GT-11** In documented wind-turbine-gearbox axial-cracking cases, subsurface crack/white-etching-area origin depths are commonly reported at under about 1 mm below the raceway surface — coincident with the depth of maximum Hertzian shear stress — concentrated on the inner races of high-speed and intermediate-speed gearbox stages; through-hardened races tend to show white-etching-crack (WEC) damage, carburized races tend to show white-structure-flaking (WSF)/spalling damage — source: Windpower Engineering & Development, "Understanding the root causes of axial cracking in wind turbine gearbox bearings" (2014); read-at-source: fetched directly in this session — "WEA slivers sit less than 0.04 inches (1 mm) below the raceway surface," reported as matching the depth of maximum Hertzian shear stress; axial cracks described as concentrated on inner rings of high-speed/intermediate-speed gearbox bearings; through-hardened vs. carburized WEC/WSF distinction as stated in the article.
- **GT-12** Full-scale dynamometer testing of wind-turbine-gearbox high-speed-shaft bearings (Guo & Keller, *Wind Energy*, 2017) found that during transient events the bearings experienced alternating periods of high stress, torque reversal, and loads below the minimum requisite to sustain pure rolling at high rotational speed, with observed indications of roller sliding; the same study found that neither rotor-moment-induced operational misalignment nor static generator misalignment measurably changed bearing loads relative to pure-torque conditions, making generator misalignment an unlikely causal factor for this damage class — source: Guo, Y. & Keller, J., "Investigation of high-speed shaft bearing loads in wind turbine gearboxes through dynamometer testing," *Wind Energy*, 2017; read-at-source: fetched directly via the NREL/OSTI abstract and key-quote record in this session — "alternating periods of high stress, torque reversals, and loads under the minimum requisite at high rotating speeds" with "indications of sliding"; "Thus, it is not likely that generator misalignment is a causal factor of WECs."
- **GT-13?** Serial-sectioning studies of white-etching-crack-affected bearing steel (Evans et al.) report that white-etching features and butterfly cracks predominantly nucleate at small non-metallic inclusions (sulfide/oxide types), that butterfly cracks can themselves propagate into WEC networks, and that the subsequent large-scale flaking (white structure flaking, WSF) is attributed by some researchers to hydrogen release/diffusion and/or transient operating conditions, with the detailed formation mechanism still disputed as of the cited reviews — source: Evans, M.-H. and co-workers, serial-sectioning WEC studies (University of Southampton), and a 2018 review (Su et al.) questioning the rubbing-generates-WEA mechanism; reported-by-delegate: identified via web search snippets summarizing these papers; the primary papers themselves were not opened in this session.

- **GT-14?** Hydrogen-content measurements on rolling-contact-fatigue-tested bearing steel (Richardson et al., *Tribology Letters*, 2018) found hydrogen diffusing into rolling elements that developed numerous WECs, with concentrations rising for longer test durations and negligible hydrogen in raceways where no WECs formed; the authors link the hydrogen's entry route to tribochemical decomposition of lubricant and/or water contamination at wear-induced nascent surfaces or heterogeneous tribofilm regions — source: Richardson, A.D. et al., "The Evolution of White Etching Cracks (WECs) in Rolling Contact Fatigue-Tested 100Cr6 Bearing Steel," *Tribology Letters*, 2018; reported-by-delegate: identified via web search snippets summarizing this paper; the primary paper was not opened in this session.
- **GT-15?** Electrically-induced bearing damage (EIBD), from stray shaft currents, common-mode converter voltage, or electrostatic discharge, is a documented failure pathway in converter-fed electric machines and wind-turbine drivetrains, producing arcing-type damage (asperity-contact pitting, local melting/re-solidification) at the rolling contact; three commonly cited current sources in converter-fed machines are electrostatic charging, magnetic-flux asymmetry, and common-mode voltage combined with fast switching slew rate; mitigations include bearing electrical insulation and shaft-grounding brushes, neither fully effective against very-high-frequency current components — source: STLE/TLT feature on electrotribology (2026); bearing-current patent literature; Chalmers University research on electrostatic-discharge-triggered main-bearing currents in wind turbines (2023, *Wind Energy Science*); reported-by-delegate: identified via web search snippets; the patents and the Chalmers paper were not opened directly in this session.
- **GT-16?** A documented sample of roughly 80 wind-turbine gearbox bearing failure records attributes the large majority (reported $_{71\%}$) to axial cracks, concentrated in the inner raceways of high- and intermediate-speed-stage bearings, with spalling ($_{}$23%) and pitting (~6%) as minority categories; laboratory rolling-sliding rig work (Argonne) reproducing this damage found that lubricant chemical additives and electrical current across the bearing surface can each contribute to axial cracking — source: Gegner, J. & Nierlich, W. (SKF), presentation at the 2011 NREL Wind Turbine Tribology Seminar; related NREL fleet-failure-record survey; reported-by-delegate: identified via web search snippets describing this presentation and survey; the primary slide deck and survey report were not opened directly in this session (the slide deck PDF was located but not fetched).
- **GT-17?** The two-parameter Weibull shape parameter ("slope") commonly used to characterize rolling-element-bearing fatigue-life dispersion under the classical Lundberg-Palmgren model is reported in the general bearing-engineering literature in the approximate range e ≈ 1.1–1.5 for ball/roller bearings — source: general bearing-engineering/tribology texts (e.g., Harris, *Rolling Bearing Analysis*); the specific exponent value most often associated with Lundberg-Palmgren theory is recalled from general engineering knowledge rather than read from a specific text or standard in this session; unverified: a precise citation-checked value was not obtained this session, so this input is carried as a bracket (e = 1.125 to e = 1.5) in chain C2 rather than as a single point figure, to absorb exactly this uncertainty.

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?-marked: GT-8, GT-13, GT-14, GT-15, GT-16, GT-17 (6 of 17)
Read-at-source: GT-1 — Facts of the Case, subsurface-origin sentence (metallographic finding)
Read-at-source: GT-2 — Facts of the Case, same sentence (crescent-shaped microcracks)
Read-at-source: GT-3 — Facts of the Case, metallographic-section sentence (WEC + butterfly
microstructure)
Read-at-source: GT-4 — Facts of the Case, disassembly sentence (axial-aligned spalling ~18 mm)
Read-at-source: GT-5 — Facts of the Case, L10/operating-hours sentences, with the 0.108 ratio
recomputed directly from the stated figures
Read-at-source: GT-6 — Facts of the Case, condition-monitoring sentence (vibration kurtosis step
increase)
Read-at-source: GT-7 — Facts of the Case, first bullet (bearing identification)
Read-at-source: GT-9 — standard lubricant viscosity–temperature relation, textbook-level
background physics
Read-at-source: GT-10 — Hertz (1881) line-contact theory, textbook-level background physics
Read-at-source: GT-11 — Windpower Engineering & Development (2014), fetched directly this
session, WEA-depth and inner-race-location quotes
Read-at-source: GT-12 — Guo & Keller, *Wind Energy* (2017), fetched directly this session via
NREL/OSTI abstract, transient-sliding and misalignment quotes
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4. Derivation Chains



Solid = read at source · dashed = not read at source (?) · fill = confidence · amber dashed = unverified input

Conclusion C1: The failure is subsurface-initiated; boundary lubrication / surface wear is not the operating mechanism

GT-1 (crack origin ~0.4 mm subsurface) + GT-8? (ISO 15243: surface distress is surface-initiated) + GT-9 (cold raises, not lowers, viscosity) + GT-10 (Hertzian line contact: max subsurface shear sits below the surface) → the cold-snap trigger the hypothesis names is physically unavailable, because falling temperature raises oil viscosity rather than lowering it → boundary-lubrication and EHD-film-breakdown damage is, by the surface-distress classification, a surface-initiated phenomenon that predicts a crack origin at the raceway surface → Hertzian line-contact theory independently establishes that a subsurface origin is the physically expected locus for fatigue-type damage under load, giving a positive alternative explanation for GT-1's depth rather than treating it only as a negative result against boundary lubrication → the measured origin sits about 0.4 mm inside the material rather than at the surface, contradicting that surface-origin prediction regardless of which precise cold-weather story is substituted for "low viscosity" *[Assumes: A-15 — the sectioned plane captured the true crack-initiation depth, not an oblique-sectioning artifact]* → both of the hypothesis's load-bearing preconditions fail independently, so boundary-lubrication wear-through is not the operating mechanism of this failure

Confidence: MEDIUM — GT-8? (the ISO 15243 surface-vs-subsurface classification) is unverified because the standard itself was not opened this session. Verification path: open ISO 15243 directly, or obtain a controlled comparative metallurgical exam confirming that boundary-lubrication-wear specimens in this bearing class show surface rather than subsurface origin. The conclusion is not sensitive to GT-8? alone, since GT-1 is a direct case measurement placing the origin subsurface regardless of which standard names the general rule, but D-07 still requires this chain's rating to reflect the `?`-marked input on its head.

Conclusion C2: Classical statistical scatter alone is a weak explanation for the magnitude of the life shortfall

GT-5 (14,000 h vs L10 130,000 h, ratio 0.108) + GT-17? (Weibull shape e ≈ 1.125–1.5 bracket) → treating L10 as the 10%-failure point of a two-parameter Weibull life distribution and solving for the scale parameter at each end of the shape bracket gives a probability of failing by 14,000 h of roughly 0.37% at the steeper end (e = 1.5) and roughly 0.86% at the shallower end (e = 1.125) → across that whole bracket, failing this early is a low-probability event under classical rolling-contact-fatigue scatter alone, on the order of one bearing in roughly 110 to 270 → classical statistical scatter in an otherwise-ordinary Lundberg-Palmgren fatigue process is therefore a weak stand-alone explanation for the observed shortfall, and either an unmodeled overload of the certified design spectrum or a non-classical degradation pathway, or both, should be sought rather than assumed away

Confidence: MEDIUM — GT-17? (the Weibull shape exponent) is unverified this session; the bracket (e = 1.125 to e = 1.5) is used precisely so the conclusion holds across the plausible range rather than resting on one unverified point value, and the result is tail-robust (even a substantially wider bracket still describes a rare event). Verification path: confirm the Lundberg-Palmgren Weibull slope for this bearing class against Harris, *Rolling Bearing Analysis*, or ISO/TR 1281-2. A rival to this chain's narrow claim — that the real operating load spectrum exceeded the certified design spectrum (A-5, unverified) — is incorporated into this chain's own conclusion ("or an unmodeled overload… should be sought") rather than left as a competing, unaddressed endpoint, so it does not independently short the Rivals axis here.

Conclusion C3: The evidence identifies the white-etching-crack / white-structure-flaking (WEC/WSF) axial-cracking failure family, not classical inclusion-only fatigue and not boundary-lubrication wear

C1 (boundary lubrication ruled out, MEDIUM) + C2 (classical-alone improbable, MEDIUM) + GT-2 (crescent microcracks) + GT-3 (WEC + butterfly microstructure) + GT-4 (axial spall ~18 mm) + GT-11 (literature depth/location match) → the subsurface origin, crescent-crack morphology, and white-etching/butterfly microstructure together form a combination that neither ordinary boundary-lubrication wear, already ruled out in C1, nor an undifferentiated "just RCF" story, already weakened in C2, predicts on its own *[Assumes: A-16 — white-etching cracks co-occurring with the butterfly, not the butterfly wing alone, is what is diagnostic of the WEC/WSF pathway rather than ordinary inclusion-driven classical RCF]* → that same combination — subsurface origin at sub-millimetre depth, axial orientation, and white-etching microstructure on a high-speed-shaft inner race — matches the documented signature of the white-etching-crack / white-structure-flaking (WEC/WSF) and axial-cracking failure family reported for wind-turbine gearbox HSS bearings → the operating failure mode is therefore a member of the WEC/WSF axial-cracking family, not classical inclusion-only Lundberg-Palmgren fatigue and not boundary-lubrication wear-through

Confidence: MEDIUM — this chain is capped at the lowest-rated chain its head cites, MEDIUM (both C1 and C2 are MEDIUM, for the reasons stated on their own confidence lines, which this line does not re-explain). GT-11 is itself read at source and would support HIGH on its own, but the ceiling rule holds regardless. A rival addressed here: contamination-indentation (a hard debris particle denting the surface and creating an ordinary subsurface stress riser) could in principle also produce a subsurface-origin crack, but it has no documented pathway to the white-etching, nanocrystalline-microstructure signature in GT-3; GT-3 is what rules this rival out, so it is not carried forward as live.

Conclusion C4: The evidence does not yet discriminate among the three live candidate triggering mechanisms for the WEC/WSF family

GT-7 (HSS generator-end position) + GT-12 (transient sliding literature) + GT-13? (inclusion/WEC literature) + GT-14? (hydrogen measurement literature) + GT-15? (EIBD literature) + GT-16? (axial-crack prevalence / lab-reproduction literature) + C3 (WEC/WSF family identified, MEDIUM) → three literature-documented triggering pathways converge on producing exactly the WEC/WSF signature identified in C3: stray electrical current / electrically-induced bearing damage at the asperity contact, hydrogen embrittlement from

tribochemical lubricant or moisture breakdown, and transient torque-reversal / skidding-driven adiabatic shear at light or variable HSS load → the bearing's generator-end position inside a converter-coupled drivetrain places it inside the recognized electrical-damage risk domain, and the HSS position independently places it inside the recognized skidding/transient-load risk domain, so neither candidate can be preferred over the other from position alone → hydrogen embrittlement is not cleanly separable from either of the other two routes, because electrical arcing and frictional or tribochemical lubricant breakdown are each independently capable of generating the hydrogen the literature implicates, so a positive hydrogen finding alone would not discriminate between them → none of the three candidate triggering mechanisms is ruled out, and none is confirmed, by the evidence available in this case record, so mechanism-level discrimination among them remains open pending the confirmatory tests named in §6

Confidence: LOW — two axes are short at once. Inputs: four `?`-marked ground truths sit on the head (GT-13?, GT-14?, GT-15?, GT-16?), none read at source; verification path is to open the Evans, Richardson, EIBD-mechanism, and Gegner/Nierlich primary sources directly. Rivals: three candidate triggering mechanisms remain live and nothing in this case record settles among them; verification path is the confirmatory test suite named in §6 (bonding/grounding conductance measurement, lubricant oil-chemistry and hydrogen-content analysis, SEM/EDS examination of the crack faces for electrical-discharge recast/melt signatures, and a slip/skidding calculation against this turbine's own SCADA torque history), each test individually capable of ruling one candidate in or out.

Conclusion C5: The operator's lubricant-viscosity / boundary-lubrication hypothesis is refuted as root cause

C1 (boundary lubrication ruled out, MEDIUM) + C3 (WEC/WSF family identified, MEDIUM) → the operator's hypothesis requires both a genuine cold-driven viscosity drop and a surface-origin wear mechanism, and C1 has already shown both of those load-bearing preconditions to be false for this failure → C3 has independently identified a different, subsurface microstructural mechanism, the WEC/WSF axial-cracking family, as the one the evidence actually matches → the operator's lubricant-viscosity / boundary-lubrication hypothesis is therefore refuted as the root cause, though lubricant condition remains a legitimate confirmatory test and a possible secondary hydrogen-source contributor rather than the primary driver

Confidence: MEDIUM — capped by C1 and C3, both MEDIUM, for the reasons stated on their own confidence lines, which this line does not re-explain. This chain's own inference step is sound given its inputs; the downgrade is entirely inherited from the cited chains.

Conclusion C6: The recommended interim action is a hedged composite plan, not a single-mechanism fix or a full stop-and-wait

C4 (mechanism undetermined, LOW) + GT-7 (bearing criticality / position context) → because C4 leaves three candidate triggering mechanisms live, any replacement plan must be scored on how well it performs across all three candidates rather than on a single assumed mechanism → scoring the three candidate plans (like-for-like replacement; full stop-and-wait for lab results; a composite interim action) against five weighted criteria — downtime cost, 12-month recurrence risk, diagnostic completeness, incremental cost, and lead time — disqualifies like-for-like replacement outright as a must-have failure, because it changes none of the three live risk drivers *[Assumes: A-17 — the stated criterion weights reflect this operator's actual risk tolerance, not just the analyst's default priorities]* → among the two remaining viable plans, the composite plan, an upgraded WEC-resistant/insulated bearing plus immediate bonding-path restoration with the full lab test suite run in parallel, scores 72 against 62 for holding the turbine down for full test completion, driven mainly by the downtime and lead-time criteria → a flip-test across every criterion shows no single criterion's weight can be moved far enough within the 1–5 scale used to reverse that ranking, so the result is robust rather than a close call → the recommended interim action is therefore the composite plan: install an upgraded WEC-resistant/insulated bearing and restore the bonding path immediately, while the full confirmatory test suite runs in parallel on the retained failed bearing and oil sample →[2nd] maintenance crews facing immediate downtime pressure may be tempted to skip the parallel lab-test and bonding-path

steps once the turbine is back online, which would leave the "diagnosis correct but not acted on" failure mode live, named and dispositioned in the adversarial pass below →[2nd] the gearbox-bearing OEM and the converter OEM each have an incentive to prefer whichever root-cause attribution is cheaper for their own warranty exposure, which can bias which confirmatory test result gets acted on first →[3rd] if the fleet-wide bonding survey this plan schedules confirms a converter-side common-mode-voltage or grounding-design issue across sister turbines, the replacement specification and maintenance procedure become a fleet-wide retrofit rather than a single-unit repair →[3rd] if instead the transient-torque/skidding pathway is confirmed as dominant, the fix is a control or torque-limiting change rather than a bearing-material change, and a bearing-only fix would keep failing on the same cadence at the next comparable grid event

Confidence: LOW — capped by citing C4 (LOW): the mechanism-discrimination uncertainty in C4 is exactly why this trade-off recommends a hedge rather than a single targeted fix, which is the correct, calibrated rating given C4′s own rating rather than a defect of this chain. The ranking among interim-action options is itself robust (the flip-test found no single-criterion reweighting reverses it), but that robustness concerns which interim action to take, not which exact mechanism is operating, so it does not raise the band above the ceiling C4 sets.

Conclusion C7: The condition-monitoring program caught this defect, and the turbine was pulled, well before any normal wear-out curve would predict a problem

GT-5 (14,000 h vs L10 130,000 h, ratio 0.108) + GT-6 (HSS vibration-kurtosis step increase at ~14,000 h triggered removal) → the step increase in HSS vibration kurtosis coincides with the same ~14,000-hour point at which the bearing had reached only about 11% of its L10 design life, a point no normal wear-out curve would flag on its own → removal was triggered by that monitoring flag rather than by a run-to-failure event, so the bearing was pulled for examination while the spall was still a single, bounded feature rather than after a secondary cascade → the condition-monitoring program performed as intended on this unit, independent of which root-cause mechanism is eventually confirmed, which is also why the evidence available for this diagnosis is as clean as it is

Confidence: HIGH — both head inputs are direct, unsuffixed case facts; each hop follows by direct deduction from the stated hours and the stated trigger event; no rival account of the monitoring system's performance is raised anywhere in this analysis, and none is apparent. This chain does not feed any later chain's head and is not relied on to establish the root-cause mechanism; it establishes only that the evidentiary record was preserved promptly, which the mechanism-identification chains (C1–C6) then make use of as a premise of having clean evidence to examine, not as a source of mechanism information itself.

Conclusion C8: The failure-surface geometry alone is the signature of a developing fatigue-type crack network, not a single sudden-overload fracture

GT-1 (crack origin ~0.4 mm subsurface) + GT-2 (crescent-shaped microcracks) + GT-3 (WEC + butterfly microstructure) + GT-4 (axial spall ~18 mm) → a single sudden-overload fracture would show a transgranular cleavage or ductile-tearing surface originating at the point of peak instantaneous stress, not a network of crescent-shaped microcracks propagating from one consistent subsurface origin *[Assumes: A-18 — standard fracture-mechanics classification distinguishes overload fracture morphology from progressive fatigue-crack-network morphology]* → a white-etching/butterfly microstructure requires many accumulated cycles of subsurface shear to develop and is not a signature any one-time overload event produces, which independently corroborates a progressive rather than sudden process → the observed combination, a localized subsurface origin, branching crescent microcracks, a white-etching/butterfly microstructure, and a bounded 18 mm spall, is instead the signature of a progressively enlarging fatigue-type crack network → the failure-surface geometry alone, independent of which specific degradation mechanism eventually gets confirmed, establishes that this was a developing fatigue-type defect rather than a one-time overload fracture

Confidence: HIGH — all four head inputs are direct, unsuffixed case facts (metallographic and disassembly findings); each hop follows from established fracture-mechanics and tribology regularities (transgranular/ductile overload morphology vs. progressive fatigue-crack-network morphology; the cycle-count

requirement for white-etching microstructure formation) rather than from any `?`-marked or assumption-dependent premise. The strongest rival, a sudden overload event, is directly ruled out by this chain's own first two hops rather than left live, which is what permits HIGH on the Rivals axis as well. This chain establishes only the overload-vs-fatigue distinction; it does not by itself identify which fatigue-type mechanism (classical, WEC/WSF, or otherwise) is operating, which is why C1 through C6 remain necessary and are not superseded by it.

5. Abandoned Reasoning

Dead End: The operator's hypothesis — low lubricant viscosity in a cold snap drove boundary lubrication and wore the race through

What was tried: The inversion technique was applied directly to the operator's stated claim. Claim: "insufficient EHD film thickness from low lubricant viscosity during a cold-snap period caused boundary-lubrication contact and wore the inner race through to failure." Inverted form: "insufficient EHD film thickness from low viscosity during the cold snap did not cause this failure." Five failure-guaranteeing conditions for the inverted form (i.e., conditions under which the original claim is false) were enumerated: (i) the cold snap in fact raised rather than lowered the oil's viscosity; (ii) the observed damage originates below, not at, the raceway surface; (iii) the crack-flank microstructure shows features (white-etching, butterflies) with no established formation pathway via viscosity thinning or asperity friction alone; (iv) the crack morphology (crescent-shaped networks) matches a published fatigue-crack morphology rather than the scuffing/polishing/micropitting morphology boundary lubrication produces; (v) the time-course and magnitude of the life shortfall (9×, over 14,000 h) do not match the time signature of a single-season viscosity-driven wear-through process. Two of these were tagged load-bearing — the claim does not survive either being false: (i) a genuine viscosity drop, and (ii) a surface-origin crack.

Why abandoned: Both load-bearing preconditions fail directly against already-established ground truths, independently of each other. Precondition (i) fails on GT-9: cooling raises, not lowers, lubricant viscosity, so the hypothesis's own named trigger has the wrong sign for a cold snap as stated — the only coherent cold-weather lubrication risk is the opposite one, cold-start starvation from oil too viscous to circulate quickly, which is a different failure story and still predicts surface-origin damage. Precondition (ii) fails on GT-1: the measured crack origin is about 0.4 mm below the raceway surface, inside the Hertzian subsurface-shear-stress band (GT-10), not at the surface where boundary-lubrication damage initiates by mechanism (GT-8?). The three corroborative preconditions independently reinforce the same conclusion: GT-3′s white-etching/butterfly microstructure has no documented formation pathway via viscosity thinning alone (chain C3); GT-2′s crescent-shaped microcrack morphology matches the WEC/WSF literature's reported crack networks (GT-11, GT-13?) rather than scuffing or polishing; and GT-5/C2′s statistical improbability of a 9× life shortfall under ordinary scatter is not the signature a slow, single-season wear-through process would leave behind on a condition-monitoring trend (a sudden step increase in kurtosis, GT-6, is more consistent with a developing fatigue defect than with progressive abrasive wear). The counter-intervention the hypothesis would prescribe — a lubricant change-out or viscosity-grade change — does not address any of the three live candidate mechanisms identified in C4 (electrical, hydrogen-tribochemical, or transient-load skidding), so acting on it would not prevent recurrence even if it were cheap to try.

What it ruled out: Lubricant viscosity and boundary lubrication as the primary cause of this specific failure, and the corresponding counter-intervention (lubricant change-out alone) as an adequate fix. It does not rule out lubricant condition as a secondary, contributing factor via the hydrogen-tribochemical pathway (C4) — that remains open pending the oil-chemistry test in §6. A skeptic re-running this diagnosis on the next HSS bearing failure in the fleet should not re-open the boundary-lubrication pathway unless that bearing's failure surface shows a genuinely surface-origin crack; if it does, this dead end's reasoning does not transfer and the diagnosis restarts from a different observation set.

Dead End: Classical Lundberg-Palmgren inclusion-only rolling-contact fatigue as the sole explanation

What was tried: Testing whether ordinary subsurface-inclusion-initiated RCF, with no non-classical mechanism at all, could explain the failure on its own — i.e., whether the butterfly microstructure and the premature life were both within the range classical fatigue alone produces, given enough scatter and ordinary inclusion content.

Why abandoned: Two independent lines of evidence weigh against sole classical RCF, though neither is individually dispositive. First, GT-3 reports white-etching cracks, not just butterflies, co-located with the subsurface origin; isolated butterflies around inclusions are a documented classical-RCF feature, but white-etching cracks are not a documented product of classical inclusion fatigue alone (A-16; this is treated as an unverified-but-literature-supported discriminator, not a certainty). Second, chain C2 shows that an 11%-of-L10 failure is a low-probability event (roughly 0.4%–0.9%) under classical Weibull scatter alone, which disfavors — without excluding — a "just unlucky, ordinary bearing" account. Neither line fully excludes classical RCF: an abnormally poor inclusion rating (A-8, unverified; no mill certificate or inclusion survey is in the case record) could independently shorten life through an entirely classical pathway, and that possibility is not yet closed. What tips this from "unresolved rival" to "abandoned as the sole explanation" rather than "ruled out entirely" is that classical RCF alone has no documented pathway to white-etching cracks specifically, so even if abnormal inclusions are later confirmed to be a contributing factor, they would not by themselves explain GT-3, and the WEC/WSF-family diagnosis (C3) would still stand as at least a co-contributor.

What it ruled out: Treating "ordinary bearing, ordinary bad luck, no non-classical mechanism at all" as an adequate stand-alone account of this failure. It leaves open, and does not rule out, an abnormal inclusion rating as a contributing amplifier alongside a WEC/WSF-family mechanism — the confirmatory inclusion survey in §6 is what would close this residual question.

Dead End: Generator-shaft misalignment as the proximate driver of roller sliding/skidding

What was tried: Testing whether static or dynamic (rotor-moment-induced) misalignment between the generator and the HSS, rather than transient torque reversal, was the load-skid driver that could explain roller sliding in this bearing.

Why abandoned: Directly contradicted by GT-12: full-scale dynamometer testing of wind-turbine HSS bearings (Guo & Keller, 2017) found that neither rotor-moment-induced operational misalignment nor static generator misalignment measurably changed bearing loads relative to pure-torque conditions, and the authors concluded that generator misalignment is unlikely to be a causal factor for this damage class. This is a read-at-source literature finding, not an assumption, so it is treated as closing this specific path rather than merely disfavoring it.

What it ruled out: Misalignment as the mechanism behind any skidding contribution to this failure. It leaves the transient torque-reversal/sub-minimum-load pathway (also reported in GT-12) as the live skidding-related candidate in C4, distinct from and not caused by misalignment.

6. Conclusion

Recommended approach — four parts, in this order:

1. Do not act on the lubricant-viscosity hypothesis: do not change the lubricant specification solely on that basis (chain C5).
2. Install an upgraded WEC-resistant/insulated replacement bearing and restore the shaft-to-ground bonding path to a verified low-conductance state immediately, before the turbine resumes generation (chain C6).
3. Run the full confirmatory test suite in parallel on the retained failed bearing, race sections, and an oil sample taken at removal: bonding/grounding conductance measurement, lubricant oil-chemistry

analysis (water, acid number, additive depletion, debris), hydrogen-content analysis and SEM/EDS examination of the crack faces for electrical-discharge recast/melt signatures, an ISO 4967 inclusion survey, and a slip/skidding calculation against this turbine's own SCADA torque/transient history (chain C4).

4. Schedule a fleet-wide bonding-ring conductance survey within the current inspection cycle for sister turbines of the same model and converter type (chain C6).

Key insight: (chain C3) The decisive discriminator was not any single exotic test but one observation the operator's hypothesis cannot accommodate: a subsurface, not surface, crack origin. Boundary-lubrication damage initiates at the contact surface by mechanism; an interior origin is structurally incompatible with that attribution regardless of how plausible the "cold snap hurt the oil" narrative sounds, and regardless of which specific cold-weather story is substituted for the one stated. Reasoning by analogy from "cold months → oil problems → bearing failure" would have led straight to the wrong intervention, a lubricant change, and would have left the two best-supported remaining drivers, a compromised electrical bonding path and transient-load-driven skidding, completely unaddressed, guaranteeing recurrence. The corollary is procedural: whenever a failure surface is available, characterizing the origin geometry before entertaining any narrative is what discriminates between entire families of candidate mechanisms in one step.

Trade-offs acknowledged: (chain C6) The recommended composite plan accepts a modest incremental cost — the insulated-bearing premium, bonding-path labor, and the lab-test panel — and defers a single, mechanism-targeted fix until the confirmatory tests return, in exchange for avoiding both the disqualified like-for-like option and the full extended-downtime option. The replacement bearing installed now is therefore a hedge against three live candidate mechanisms, not a confirmed fix for one of them (chain C4); if the tests eventually show the electrical and skidding pathways are both inactive and only a lubricant-chemistry/hydrogen pathway is live, part of the insulated-bearing premium will have been unnecessary. This is accepted deliberately because the flip-test showed the interim-action ranking is robust to a wide range of relative criterion weighting (chain C6), so paying this modest premium against unresolved mechanism uncertainty is a considered trade, not an oversight. The analysis also accepts that four of the ground truths feeding the mechanism-discrimination chain (GT-13?, GT-14?, GT-15?, GT-16?) rest on literature synthesis rather than on primary sources opened directly in this session; no part of this conclusion treats any single one of them as individually load-bearing without the others, and C4's LOW rating (chain C4) already reflects that honestly rather than hiding it inside a higher-sounding band.

Confidence: LOW — the section's rating is bound by the lowest chain it rests on: C4 and C6 are both LOW. C4 is LOW because two axes are short at once (four `?`-marked ground truths on its head, and three live, unsettled candidate triggering mechanisms); C6 is LOW because it is capped by citing C4. C3 and C5 are MEDIUM, each capped by citing C1/C2, which are MEDIUM because GT-8? and GT-17? are unverified `?`-marked inputs with named verification paths already stated on their own confidence lines (not re-explained here). The practical split is this: refuting the operator's lubricant-viscosity hypothesis (C5, MEDIUM) and identifying the WEC/WSF failure family (C3, MEDIUM) rest on comparatively solid ground, a direct metallographic measurement (GT-1) and one literature source read directly at source this session (GT-11), while determining exactly which of the three candidate triggering mechanisms is operating, and therefore exactly how to weight the interim replacement plan (C4, C6, both LOW), remains genuinely open pending the confirmatory tests named above. Neither Criterion 5 EXCEPT clause applies: no chain here is marked speculative, and none is an absent-fails derivation.